

Multi Exposure Trigger Mode

The data except this specification conform to those of each product.

1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

The function of multi exposure in one frame at Global shutter is described in this document.

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3. Features

◆ Multi Exposure Trigger mode

Support the number of times of exposure from 2 to 4095.

◆ Support mode

Sequential Trigger mode

Fast Trigger mode

* Normal mode is not supported.

◆ Supported readout drive mode

All-pixel scan mode

Vertical / Horizontal 1 / 2 Subsampling mode

2 × 2 FD binning mode (The product with a color filter is excluded)

ROI mode

Overlap ROI mode

◆ The monitor output for Exposure period

◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series / IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

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4. Register Map

The register map for Multi Exposure Trigger mode is shown below.
Please refer to the product specification for register setup other than those list.

Registers corresponding to Chip ID = 02h in Write mode. (I²C:30**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
10h	3010h	0	XMSTA [0]	Setting of master mode operation 0: Master mode operation start 1: Master mode operation stop	1	01h	I
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—

Registers corresponding to Chip ID = 04h in Write mode. (I²C:32**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
0Ch	320Ch	0	EXP_CNT_RESET [0]	Reset the number of times of exposure at Multi Exposure Trigger mode Refer to "Reset the number of times of exposure"	0	00h	I
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
0Eh	320Eh	0	EXP_MAX [11:0]	LSB	001h	01h	V
		1					
		2					
		3					
		4		Set the number of times of exposure at Multi Exposure Trigger mode			
		5					
		6		Setting range:2 to 4095			
		7		Only Fast trigger mode can be set			
0Fh	320Fh	0	EXP_MAX [11:0]	MSB	0	00h	—
		1					
		2					
		3					
		4		Fixed to 0			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

Registers corresponding to Chip ID = 06h in Write mode. (I²C:34**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
00h	3400h	0	TRIGMODE [2:0]	Global shutter mode setting 0h: Normal mode 1h: Sequential Trigger mode 2h: Fast Trigger mode Others: Setting prohibited	0h	00h	I ^{*1}
		1					
		2					
		3	TRIGTIMING [1:0]	Control of the exposure timing Only in 1d	0		I ^{*1}
		4					
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
7		Fixed to 0	0	—			
36h	3436h	0	TOUT0SEL [2:0]	TOUT0 pin setting 0h : Low fixed 1h : Exposure period monitoring	0h	00h	S
		1					
		2					
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
3Ch	343Ch	0		Fixed to 0	0	C0h	—
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4	SYNCSEL [1:0]	XHS, XVS pin setting 0h: Normal Output 3h: Hi-Z Others: Setting prohibited	0h		S
		5					
		6		Fixed to 1	1		—
		7		Fixed to 1	1		—

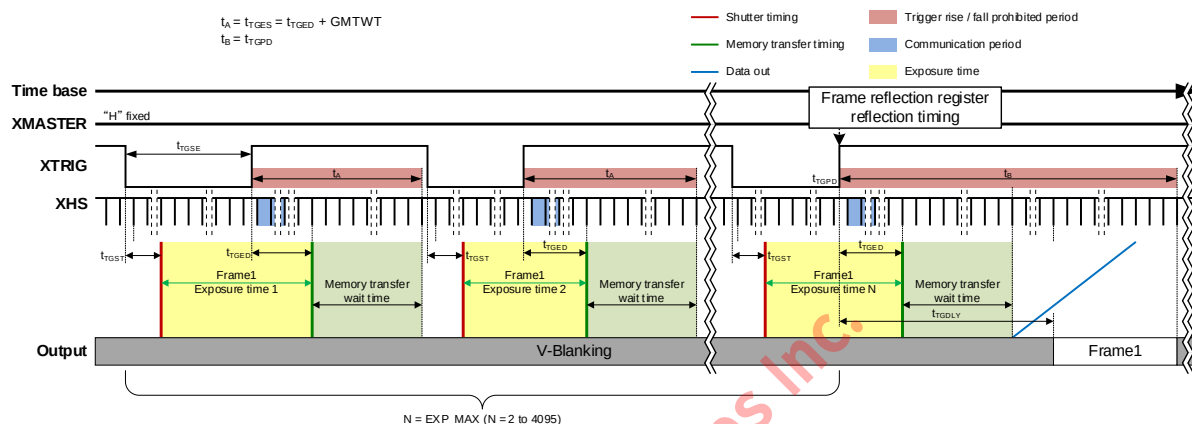
*1 Refer to "Mode Transitions of Global Shutter Operation" of DataSheet

5. Exposure Timing

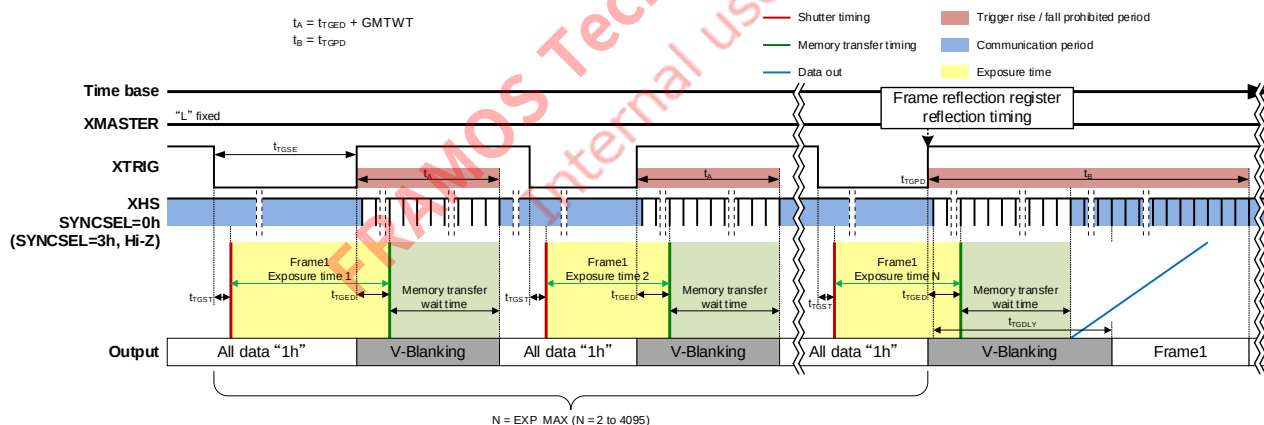
Multi exposure timing is as below.

This sensor exposes the value of EXP_MAX (Chip ID = 04h Address = 0Eh, 0Fh), and the data of the frame is output after the last exposure. During data output period, a start (fall input of XTRIG) is possible by the exposure of the next frame, but it is not possible to expose it, and to be finished (start input of XTRIG). The reflection timing position for registers marked "V" is the same as the timing of the Datasheet. However, during Multi Exposure Trigger mode, it is reflected by the trigger of the last exposure.

Refer to the Datasheet about the spec of each timing



Exposure timing in the case of EXP_MAX = N (Sequential trigger mode, Slave mode)



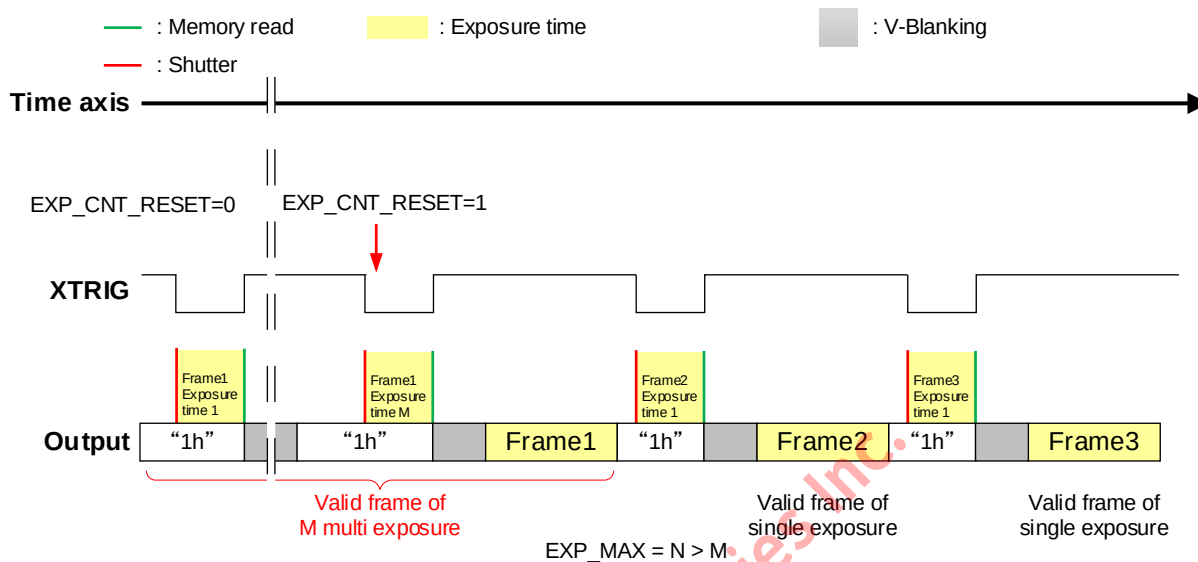
Exposure timing in the case of EXP_MAX = N (Fast trigger mode, Master mode)

6. Reset the number of times of exposure

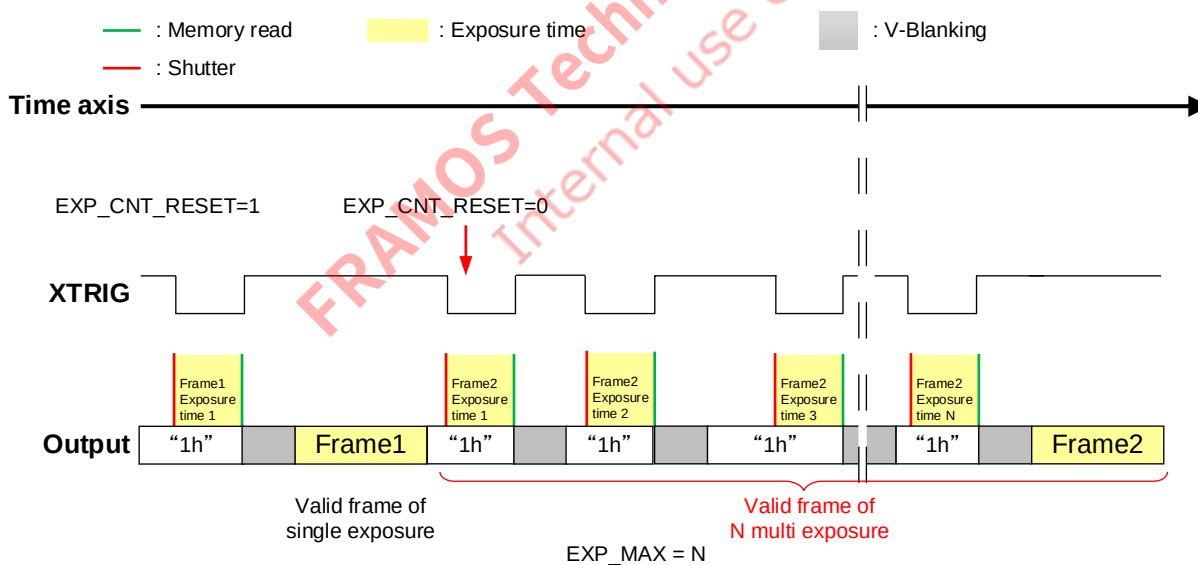
During the Multi Exposure Trigger mode, The number of times of exposure which is set by EXP_MAX can be reset.

EXP_CNT_RESET = 0 → 1: The condition becomes Single Exposure mode

EXP_CNT_RESET = 1 → 0: The condition becomes Multi Exposure mode



In the case of EXP_CNT_RESET = 0 → 1 (Fast trigger mode)



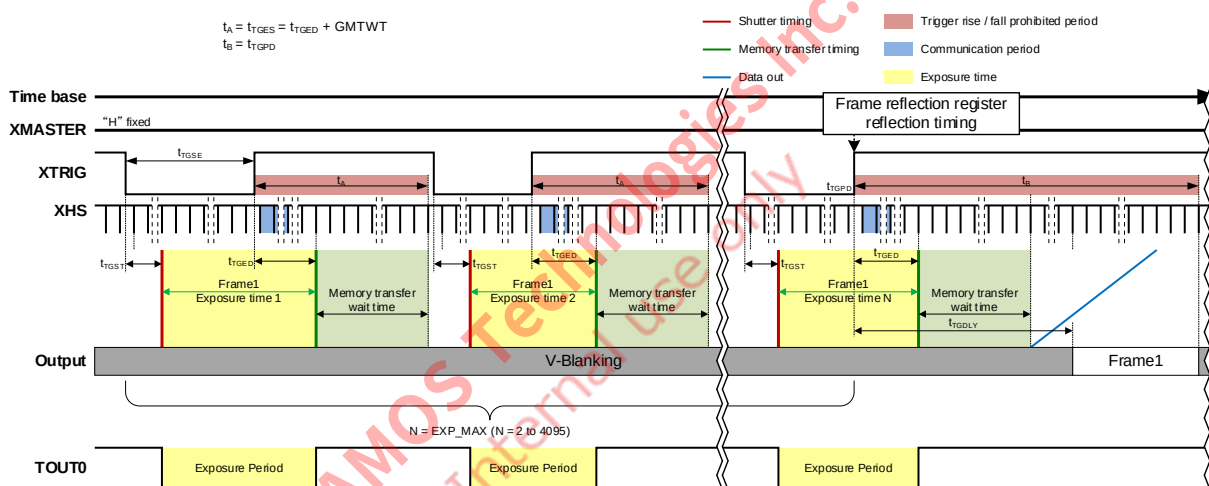
In the case of EXP_CNT_RESET = 1 → 0 (Fast trigger mode)

7. The monitor output for Exposure period

As for this sensor, TOUT0 can output the status of the exposure control signal in Multi Exposure Trigger mode.

7.1. List of registers

Register	Register details			Initial value	Description
	Chip-ID	Address (I2C)	bit		
TOUT0SEL [2:0]	06h	36h (3436h)	[2:0]	0h	TOUT0 pin setting 0h : Low fixed 1h : Exposure period monitoring



Exposure period monitor output of the Multi Exposure Trigger mode
(Sequential trigger mode, Slave mode)

8. Mode Transitions

The transition related to Multi Exposure Trigger mode is shown below. If perform the transition which is not described in table below, set via sensor standby.

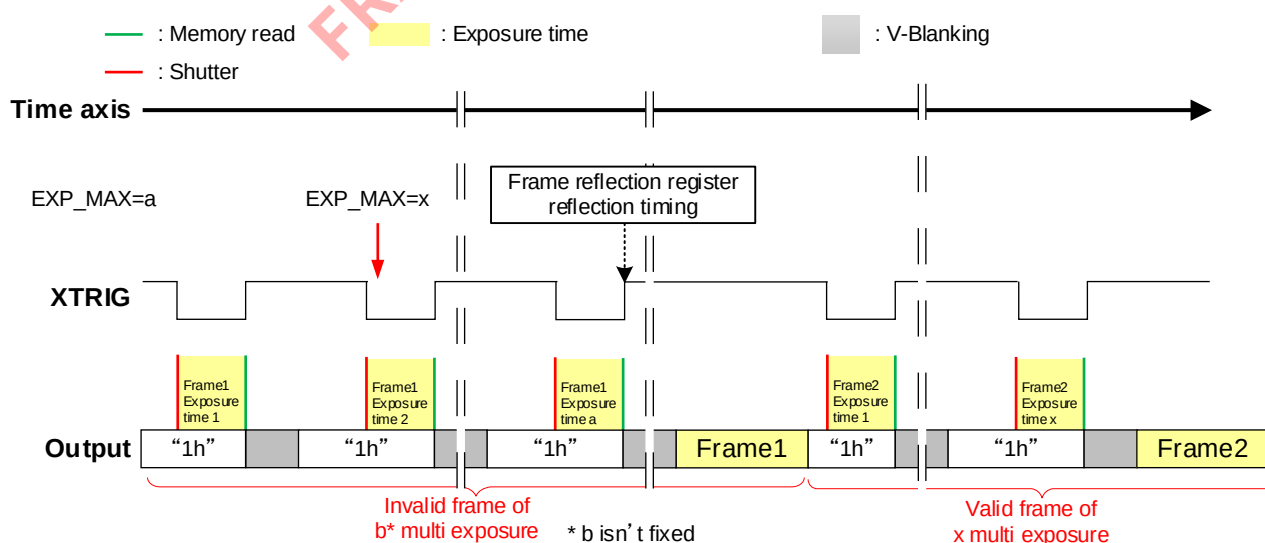
8.1. Mode Transition

Mode Transition			Remarks
Sequential Trigger mode	⇔	Multi Exposure Sequential Trigger mode	There comes out invalid frames.
Fast Trigger mode	⇔	Multi Exposure Fast Trigger mode	There comes out invalid frames.
Multi Exposure Sequential Trigger mode	⇔	Multi Exposure Fast Trigger mode	Reset the sensor (XCLR = 0).
Multi Exposure Sequential Trigger mode	⇔	Normal mode (Sensor Slave mode)	Set via the sensor standby. * Normal mode (Sensor Master mode) is limited to sensor slave. Sensor reset is required for transition to sensor master.
Multi Exposure Fast Trigger mode	⇔	Normal mode (Sensor Master mode)	Set via the sensor standby. * Normal mode (Sensor Slave mode) is limited to sensor master. Sensor reset is required for transition to sensor slave.
Change the number of times of Multi Exposure			There comes out invalid frames.
Multi Exposure Sequential Trigger mode	⇔	Multi Exposure Sequential Trigger mode	
Multi Exposure Fast Trigger mode	⇔	Multi Exposure Fast Trigger mode	

As one example, in the case of changing the number of times of Multi Exposure, it is shown as follows.

◆ Transition from normal ROI mode to Multi Frame ROI mode

The transition setting frame becomes invalid.



The reflection timing of changing EXP_MAX (Fast trigger mode)

9. Notes On Handling

Notes On Handling of the Multi Exposure Trigger mode.

- ◆ Multi Exposure Trigger mode and Dual ADC (without combination / with built-in combination) come by combination drive. (only IMX530 series and IMX535series)
- ◆ The drive with the Normal mode (the XVS/XHS same period) is non-support.
- ◆ Driving with the combination of "1 / 2 Subsampling mode" and "ROI mode" is not supported.
- ◆ Driving with the combination of "1 / 2 Subsampling mode" and "Overlap ROI mode" is not supported.

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Revision History

Version	Date	Page	Remarks
Rev.0.1	2019 / 4 / 26	–	First Edition
Rev.0.2	2019 / 6 / 28	1, 2, 7	Correction: Function name Exposure time → Exposure period
		9	Correction: Function name Dual ADC (without combination / with built-in combination)
Rev.1.0	2020 / 1 / 7	–	First Edition (Official version of IMX530 series / IMX540 series)
		1	Added: Product types (IMX531, IMX532, IMX540, IMX541, IMX542) • Correction: “conform to ... IMX530” → “conform to ... each product” • Added: Chapter “Product type” Correction: 2x2 FD binning mode • 2x2 Vertical FD binning mode → 2x2 FD binning mode • Added: “The product with a color filter is excluded” Added: SLVS output (2 ch) in the supported interface
		5, 7	Correction: Figures of Exposure timing and Exposure period monitor output • Memory wait time → Memory transfer wait time • $t_A = t_{TGED} + GMRWT + GMTWT$ → $t_A = t_{TGED} + GMTWT$ • Communication period • Added: Ellipsis marks • The start and end positions of Frame1 were shifted to the positions synchronized with the XHS. • (In the case of Fast trigger mode) The start position of t_{TGDLY} was corrected to the position of the first XHS pulse after XTRIG rising.
Rev.2.0	2020 / 12 / 17	–	Added: IMX535 series / IMX545 series / IMX565 series
		2	Added: CSI-2 serial data output
		4	Correction: Default value after reset (By register) of XMSTA with Registers corresponding to Chip ID = 02h • 0 → 1
		5	Correction: Default value after reset (By address) of TRIGMODE and TRIGTIMING with Registers corresponding to Chip ID = 06h • 04h → 00h • TRIGTIMING [4:3] → TRIGTIMING [1:0]
Rev.3.0	2021 / 3 / 4	1, 2	Added: IMX548 series / IMX568 series
		10	Added: Definition of series for Dual ADC combination drive Added: Two items about combination drive to “9. Notes On Handling”